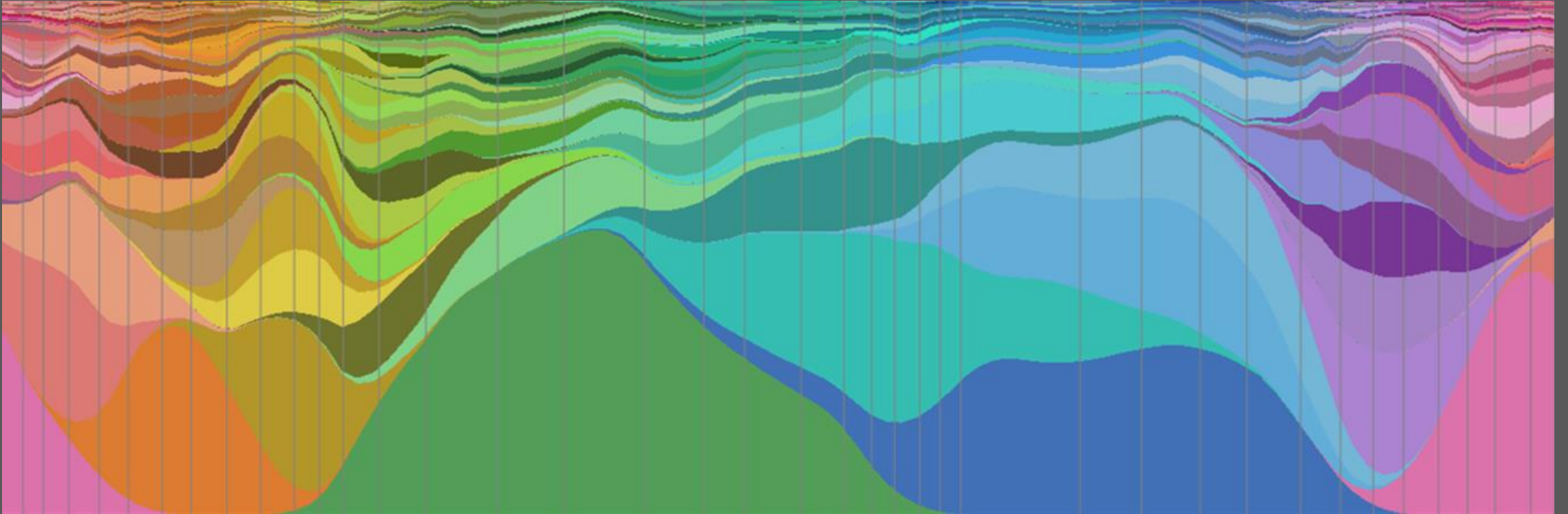


LDSCI5209 - Information Presentation and Visualisation

Week 13: State of the art visualisations



Dimitris Mylonas

Northeastern University London

Recap: Evaluation and storytelling with data

- Threats and validation
- Usability studies
- How to tell a visual story with data
- How to give a talk

Plan for today

State of the art visualisations

- Accessing literature on data visualisation (IEEE VIS, CHI)
- Critique of contemporary visualisation topics
- How to write a scientific paper
- Case study

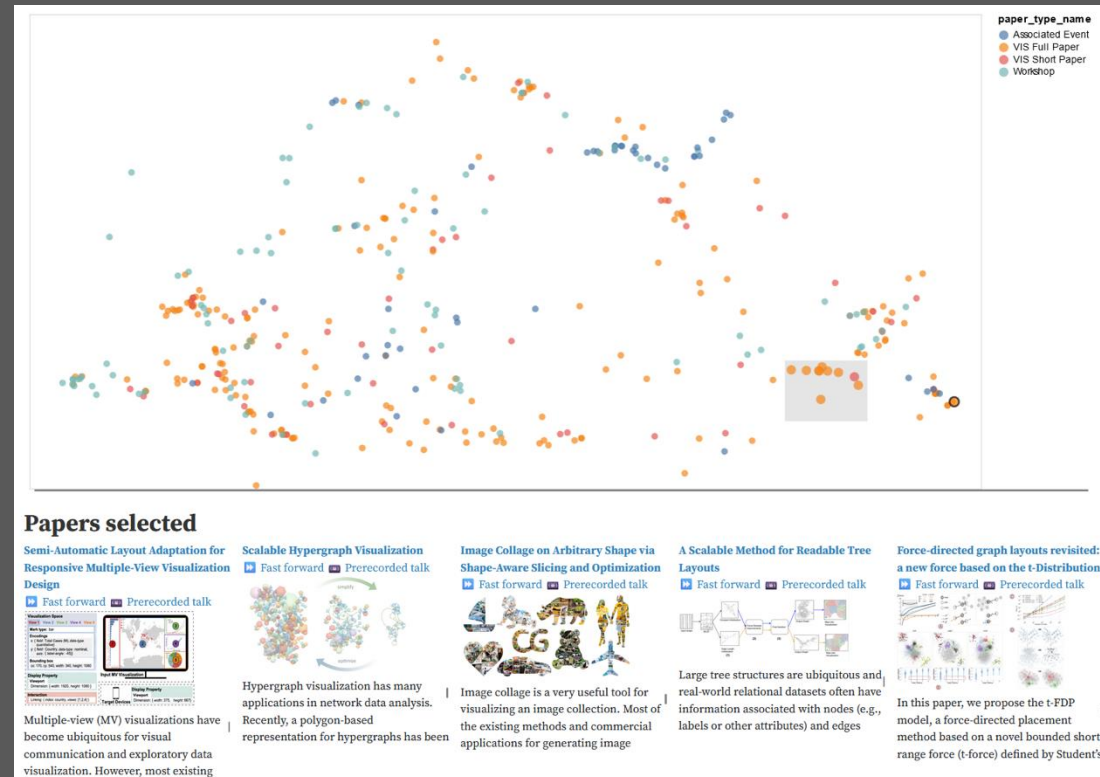
Visualisation research landscape

Key conferences and journals, libraries, trends

Key venues in data visualisation research

- **IEEE VIS:** <https://ieevis.org/>
- **ACM CHI:** <https://chi2025.acm.org/>
- **EUROVIS:** <https://www.eurovis2025.lu/>
- **Journal:** [IEEE Transactions on Visualization and Computer Graphics](#)

Review the program for IEEE VIS



<https://observablehq.com/@john-guerra/ieeevis-2023-papers>

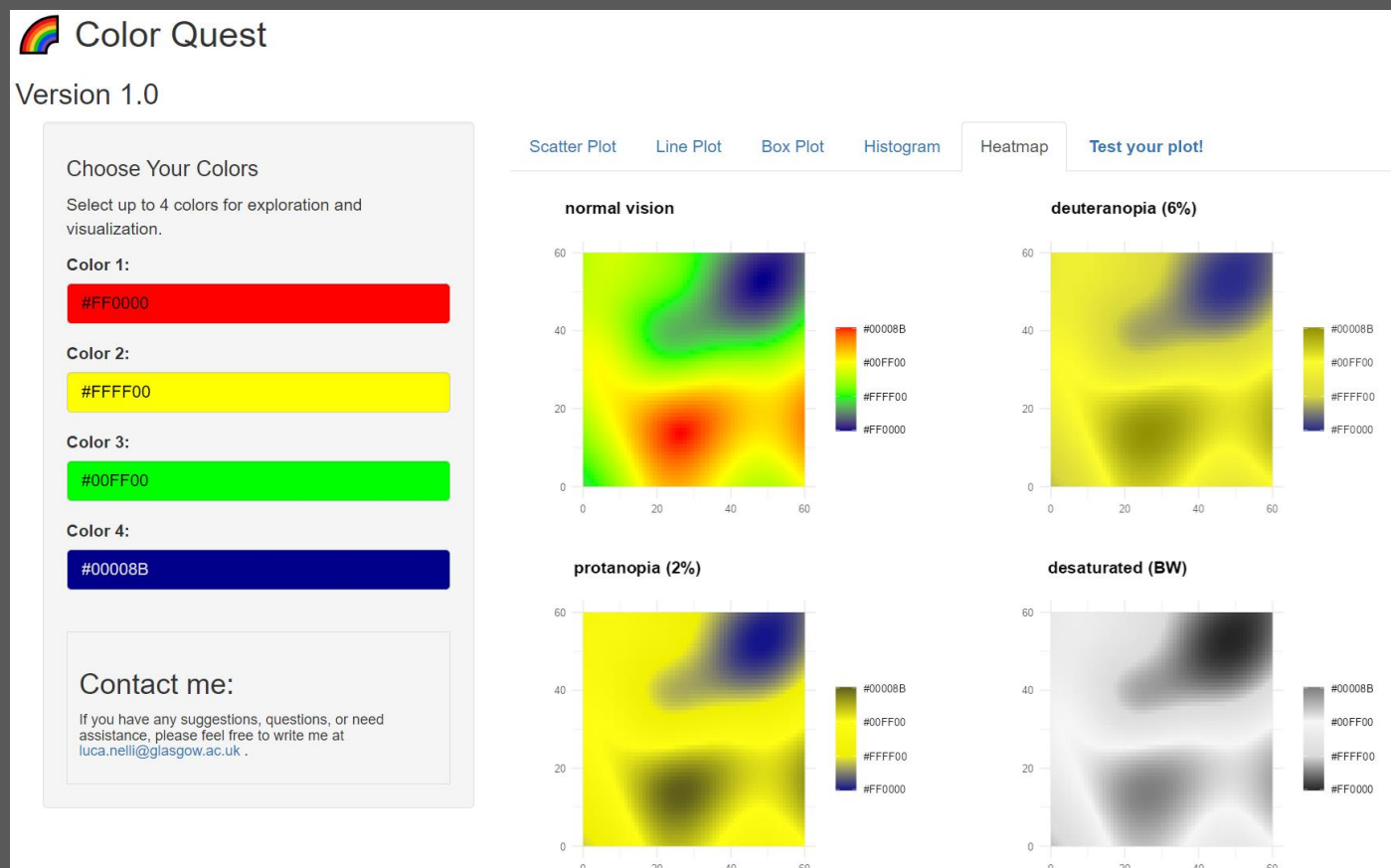
How to access and read papers

- IEEE Xplore: <https://ieeexplore.ieee.org/Xplore/home.jsp>
- ACM Digital Library: <https://dl.acm.org/>
- Google Scholar: <https://scholar.google.com/>
- Open access repositories: <https://www.arxiv.org/>
- Research groups and labs: <https://idl.uw.edu/>
<https://vis.khoury.northeastern.edu/>
- Reference manager: <https://www.zotero.org/>
<https://www.mendeley.com/>

Current trends in data visualisation

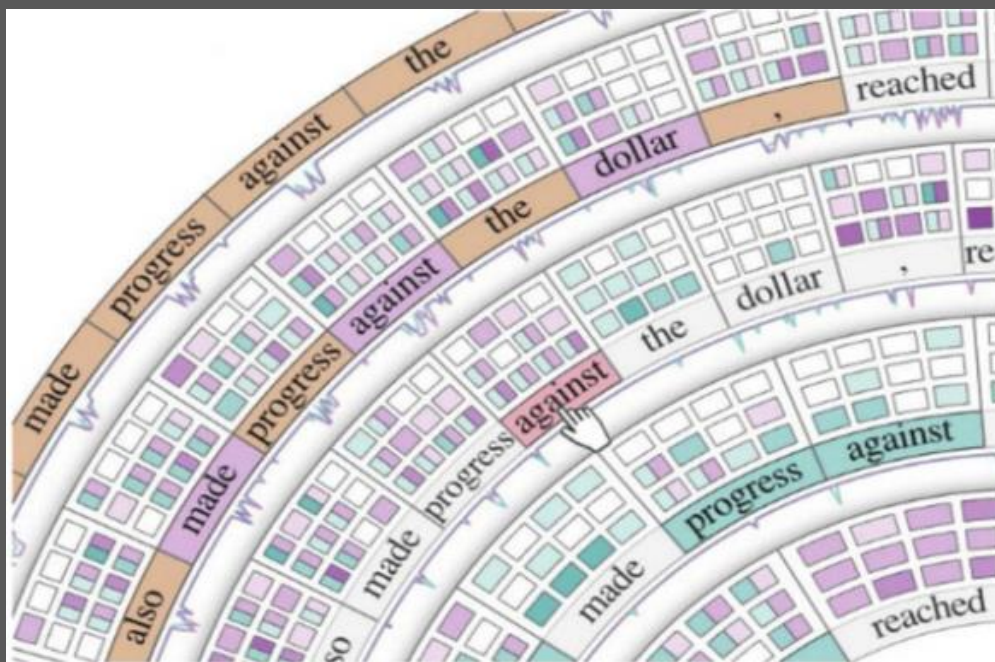
- **Accessibility and Inclusive Design:** i.e. visual impairments
- **AI Integration and Explainability:** i.e. neural networks interpretation
- **Perception and Cognition:** i.e. cultural differences, cognitive load
- **Interactive Techniques:** i.e. AR/VR, multimodal, collaborative
- **Authoring systems:** i.e. NLP interfaces, visual specification
- **Emerging Application Domains:** i.e. health, bioinformatics

Accessibility and Inclusive Design

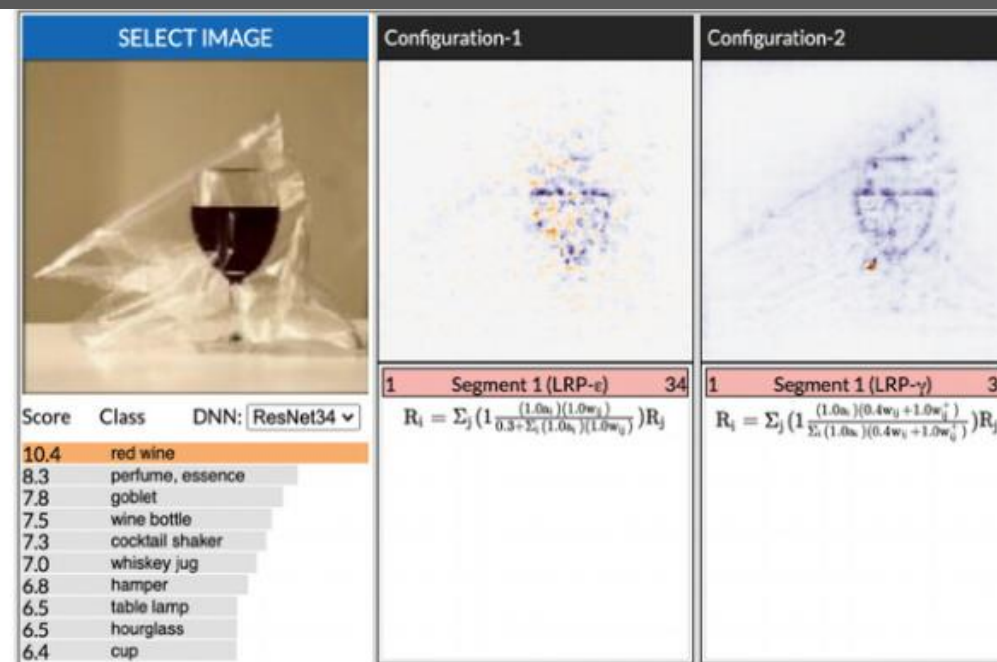


Colour Quest (Nelli, 2024)

AI Integration and Explainability

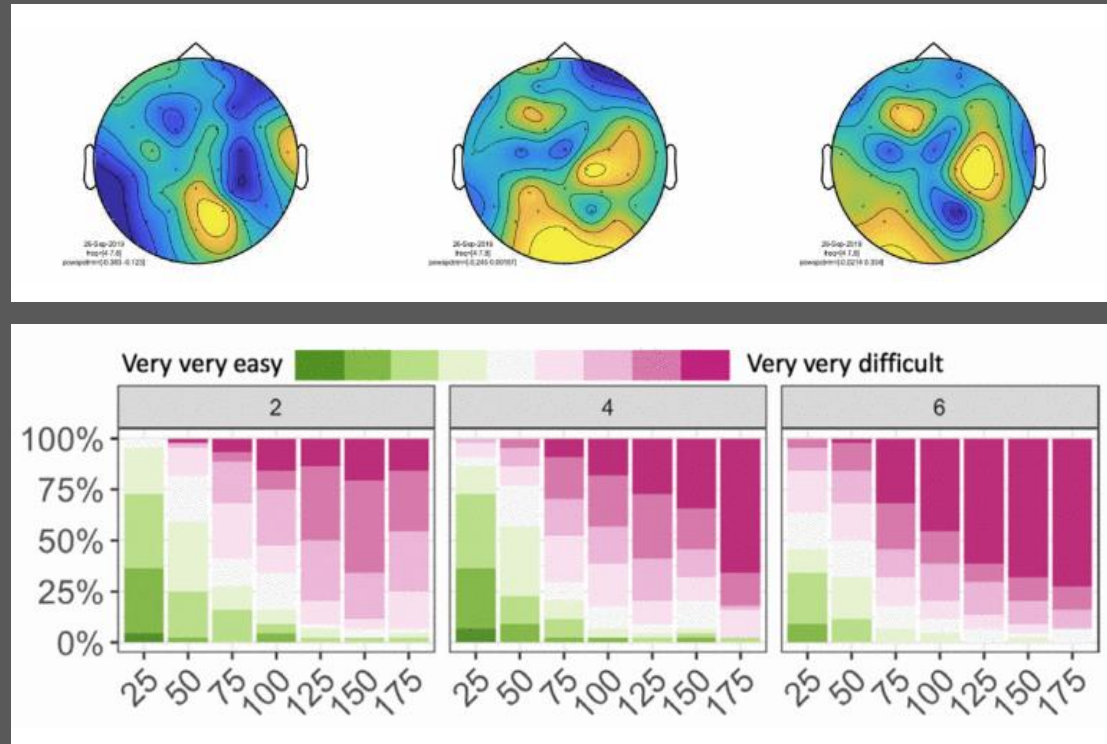


(a) Attention Flow [DWB21]



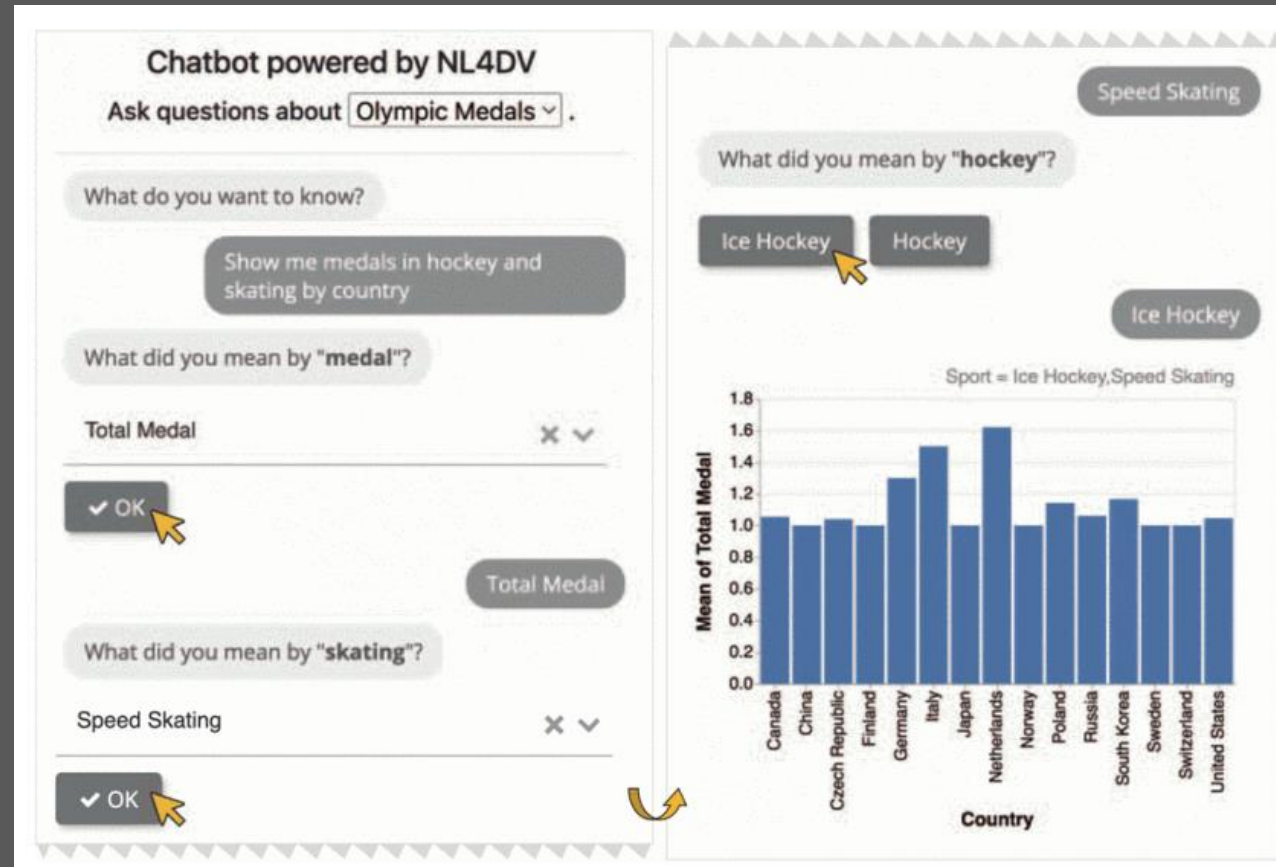
(b) VisLRPDesigner [HJZ*21]

Perception and Cognition



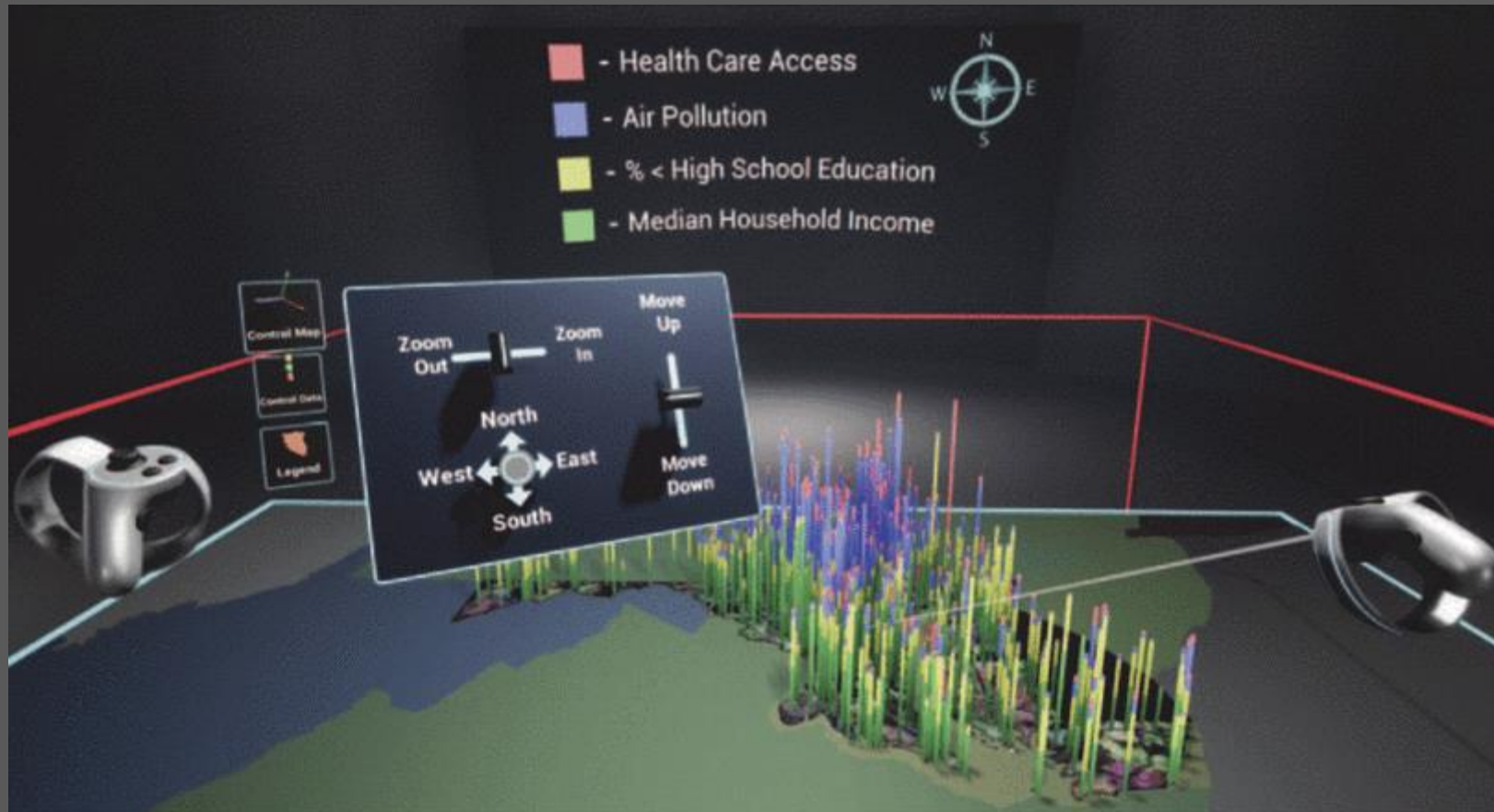
Scalability of Network Visualisation from a Cognitive Load Perspective (Yoghourdjian et al., 2020)

Authoring systems



Facilitating Conversational Interaction in Natural Language Interfaces for Visualization (Mitra et al., 2022)

Interactive Techniques (VR)



Immersive Health Data Visualization in Virtual Reality (Rajasagi, 2023)

Critical reading of research literature

Why, types, evaluation questions

Why Critique matters

- Building research and writing skills
- Identifying strengths and limitations
- Developing your own research ideas
- Understanding the field's evolution



Types of Critique: Technical, methodological, theoretical, practical

Levels of reading

- **Skimming:** get the big picture (abstract, figures, conclusion)
- **Close reading:** understand methods, results, argument
- **Critical reading:** ask questions, challenge claims, reflect

Critical reading checklist

- ☐ **Purpose:** What is the main question or goal?
- ☐ **Contribution:** What does the paper add that's new?
- ☐ **Methodology:** Is the method appropriate and rigorous?
- ☐ **Evidence:** Are claims supported by data or citations?
- ☐ **Clarity:** Is the argument logical and well-structured?
- ☐ **Bias & Limitations:** What's missing or underexplored?

Structure of scientific papers

Components, writing tips, visualisations, publication

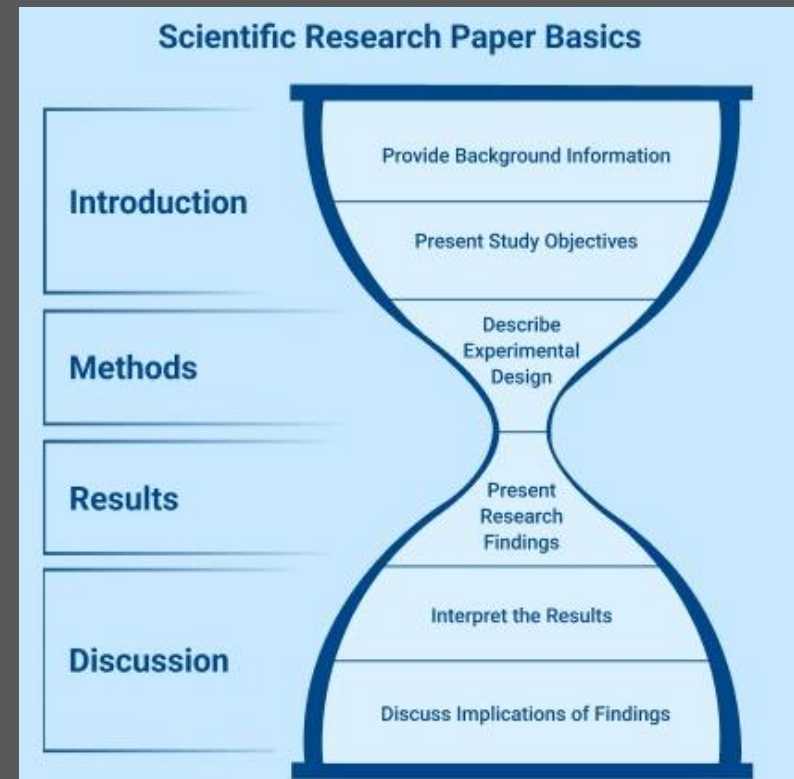
Why write scientific papers?

- Academic contribution
- Professional development
- Research community
- Real world impact



Main components of a scientific paper

- Abstract
- Introduction (why?)
- Methods (how?)
- Results (what?)
- Discussion / Conclusions (so what?)
- References



What to look for each section

Abstract

Does it accurately summarise the paper?

Introduction

Is the problem clearly stated? Why is it important? Are sources relevant and recent? Are gaps clearly identified?

Methods

Is it reproducible? Are the methods suitable for the question?

Results

Are findings clearly presented and analysed?

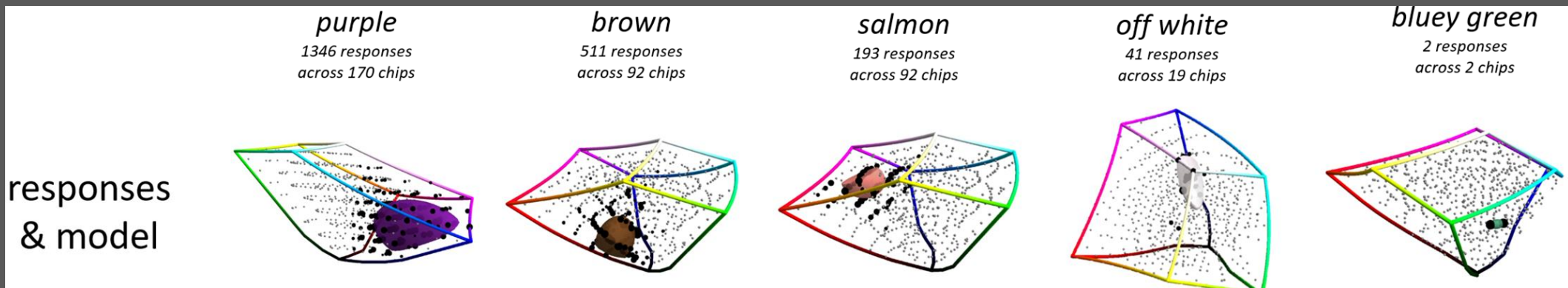
Discussion / Conclusion

Are interpretations reasonable? Are limitations discussed? Does it reflect on contributions and future work meaningfully?

Visual communication in papers

Figure Design Best Practices:

- Clear captions and references in text
- Visual hierarchy and annotation
- Use of colour and visual encoding



Categorical colour geometry (Griffin & Mylonas, 2019)

Tips for writing

- Write early and revise often
- Clear, concise language
- Focus on contribution
- Show, don't tell (use visuals!)
- Get feedback from peers/supervisors

Common pitfalls

- Poor structure
- Not stating contributions clearly
- Overloading with jargon
- Underexplaining methods or evaluation

Publication process

- Original research work
- Prepare manuscript against guidelines
- Conference deadlines and journal process
- Peer review process
- Revision and rebuttal
- Decision
- Final publication

Additional resources

How to create compelling scientific data visualizations:

<https://www.nature.com/articles/d41586-023-03393-9>

How to write a great research paper:

<https://simon.peytonjones.org/great-research-paper/>

Questions?

Case Study: Color Buddy

The **edit plane** supports drag and drop, multi-selection, and other common vector graphics operations.

The lint evaluation section surfaces checks about the current color palette.

Each check can be **ignored** if it is not relevant to the current usage context.

Every lint is also equipped with a **fixer**, generated by AI or heuristically.

Lints can also be **defined or modified** by the end user, such as increasing the max allowed colors or requiring that the first palette color is a brand color.

This panel enables manipulations of the current color palette, such as generating new colors

Examples provide a visual reference for the effect of the current palette, either through generic graphics (left) or configurable examples (right).

Color Buddy interface details:

- Top Bar:** PALETTE, MANAGE, BROWSE, CURRENT PALETTE, EXAMPLES, COMPARE, EVAL.
- Left Panel (A):** CONTROLS, COLORS. Includes "Add Color" (with search), "Adjust selected colors" (Lighten, Darken, Saturate, Desaturate, Convert To Opposing, Clip to gamut, jitter), "Distribute" (horizontal, vertical, in L space), "Align" (Left, Right, Top, Bottom, L-Min, L-Max), "Rotate" (0 to 90 degrees), "Interpolate" (Color Space: lab, Interpolation steps: 1), "Preview" (Add points), and "Current Colors" (Include quotes).
- Center Panel (B):** "Current Pal: category10", "Type: elegant", "Color Space: LCH", "Background: #0000". It features a CIE L*a*b* diagram with points and a text box: "Colors that are out of gamut are marked with an X". Below is a "CVD simulation" showing simulated colors for tritanopia. At the bottom, there are "Change example" and "Switches" for various color schemes.
- Right Panel (C):** "REGULAR", "COMPACT", "LINT-CUSTOMIZATION", "SETTINGS". It lists various accessibility checks with status indicators (e.g., "WCAG Contrast: AA (text colors only)", "WCAG Contrast: AAA (text colors only)", "CVD: Deuteranopia Friendly", "CVD: Protanopia Friendly", "CVD: Tritanopia Friendly"). Each check has a "fixer" button.
- Bottom Right (D):** "Examples" section showing "VISUALIZATIONS (VIA VEGA)" with four examples: "Example Area Chart", "Example Bar Chart", "Example Heatmap", and "Example Map".

Mixing Linters with GUIs: A Color Palette Design Probe (McNutt, 2024)

□ Purpose

What is the main question or goal?

- **Research question:**

Can automated critique (like linters) be effectively integrated into creative GUI tools for colour palette design?

- **Goal:**

Explore how designers respond to and interact with linter-style feedback during creative work

□ Contributions

What does the paper add that's new?

- Color buddy: <https://color-buddy.netlify.app/>
 - Design probe combining GUI palette design with real-time linter feedback.
 - Novel interface prototype and qualitative insights into designer behaviour.
 - Flexible or adaptive linters as a future direction.

□ Methodology

Is the method appropriate and rigorous?

- Uses a design probe, a common exploratory method in HCI.
- 5 participants, semi-structured interviews, thematic analysis.
- Interface built using a hybrid visual + linter feedback model.

□ Evidence

Are claims supported by data or citations?

- Direct participant qualitative interviews.
- Observations of user interaction.
- Clear thematic findings (e.g., “hacking the linter”).
- Citations support background and related work.
- Lacks quantitative data

□ Clarity

Is the argument logical and well-structured?

- Well-organised:
background > probe > study > findings > implications.
- Easy to follow even if you're not a colour theory expert.
- Uses visuals and examples to ground abstract points.
- The core argument is balancing automation and designer agency.

□ Bias & Limitations

What's missing or underexplored?

- Small sample size (n=5).
- Participants mostly design-savvy.
- Short-term usage only.
- Accessibility-focused use cases (e.g., colourblind users?).
- Broader user types (e.g., beginners, developers, analysts).
- Potential for misusing or over-trusting the linter.

Questions?

Assignments

- **Formative:**

Critique of a contemporary data visualisation research paper